

Week 11 – Chapter 11

MSDS 535

Overview and Objectives

- Understand Outlier detection basic concepts.
- Comprehend Statistical approaches.
- Understand Proximity-based approaches.
- Understand Reconstruction-based approaches.

What are outliers?

- An outlier is a data object that significantly deviates from the rest of the data.
- "Normal" or expected data is used to describe data objects that are not outliers, while "abnormal" data refers to outliers.
- Outliers are not the same as noisy data, as noise is random error or variance and should be removed before outlier detection.

What are outliers?

- In outlier detection, it's important to justify why the detected outliers were generated by different mechanisms than the rest of the data, often by assuming characteristics of the rest of the data and showing that outliers violate those assumptions.
- Outlier detection is related to novelty detection, as both involve identifying unusual data points, but novelty detection typically incorporates new topics or trends into the model of normal behavior once confirmed, while outliers remain distinct.

Types of outliers

- Outliers can be classified into three categories: global outliers, contextual (or conditional) outliers, and collective outliers.
- Global outliers deviate significantly from the rest of the data and are sometimes called point anomalies.
- Most outlier detection methods aim to find global outliers.

Types of outliers

- Contextual outliers deviate significantly within a specific context or with respect to certain contextual attributes.
- Contextual attributes define the object's context, while behavioral attributes evaluate whether the object is an outlier in that context.
- In contextual outlier detection, whether an object is an outlier depends on both behavioral and contextual attributes.

Types of outliers

- Contextual outliers are a generalization of local outliers and allow for examining outliers in different contexts.
- Collective outliers refer to a subset of data objects that, as a whole, deviate significantly from the entire data set, even though individual objects may not be outliers.

Types of outliers

- Collective outlier detection requires background knowledge of the relationship among data objects, such as distance or similarity measurements.
- A data set can have multiple types of outliers, and an object may belong to more than one type of outlier.
- Different types of outlier detection may be used in various applications or for different purposes, with global being the simplest.

Challenges of outlier detection

- Difficulty in effectively modeling normal and outlier behavior due to the complexity of defining normality.
- The unclear boundary between normal and abnormal data, leading to methods that assign labels or outlier scores.
- Application-specific requirements for choosing similarity measures and models make universal outlier detection methods impractical.

Challenges of outlier detection

- Dealing with noise in data is challenging as it can blur the line between normal and outlier data.
- Interpretability is a concern as users may want explanations for detected outliers, which can be addressed through methods like statistical justification.

Outlier detection methods

- Outlier detection methods are categorized in two orthogonal ways:
 - Whether expert-labeled data is provided: supervised, semisupervised, or unsupervised methods.
 - Assumptions about normal vs. outlier data: statistical, proximity-based, and reconstruction-based methods.

Outlier detection methods

- Supervised methods rely on expert-labeled data for building classification models to recognize outliers.
- Unsupervised methods assume that normal data objects exhibit patterns more frequently than outliers.
- Semisupervised methods are used when only some labeled data is available, utilizing labeled normal objects to train models for detecting outliers.

Outlier detection methods

- Statistical methods assume that normal data follows a statistical model, and deviations from the model are considered outliers.
- Proximity-based methods label objects as outliers if their nearest neighbors are significantly farther in feature space compared to most other objects.
- Reconstruction-based methods are based on the idea that normal data can be represented more succinctly than outliers, and outliers are those that cannot be well reconstructed.

Statistical approaches

- Statistical methods for outlier detection make assumptions about data normality based on generative models.
- Normal objects are found in high-probability regions of the model, while low-probability regions contain outliers.
- These methods aim to learn generative models from data and identify low-probability objects as outliers.

Statistical approaches

- Statistical methods are divided into two categories: parametric (assume a specific distribution with parameters) and nonparametric (determine the model from data).
- Nonparametric methods, like histogram and kernel density estimation, allow flexibility in the number and nature of parameters.

Parametric methods

- Parametric methods are practical for outlier detection.
- For univariate data, a common approach is assuming normal distribution, estimating parameters (mean and standard deviation), and identifying points with low probability as outliers.
- Various univariate methods include using 3σ as a threshold, Grubb's test, and the boxplot method.

Parametric methods

- Multivariate data can be handled using techniques like Mahalanobis distance or χ^2 -statistic.
- For complex data distributions, a mixture of parametric distributions can be assumed, and the EM algorithm helps learn parameters for each cluster, where objects not belonging to any cluster are considered outliers.
- Constraints on the distribution generating outliers can improve detection, for example, by assuming a larger variance for outliers.

Nonparametric methods

- Nonparametric outlier detection learns the "normal data" model from input data without pre-assumptions, offering flexibility in various scenarios.
- Histograms are a common nonparametric model for outlier detection, involving histogram construction and outlier detection based on bin assignments.
- Choosing an optimal bin size for histograms can be challenging, leading to false positives or false negatives.

Nonparametric methods

- Kernel density estimation estimates data's probability density function, facilitating outlier detection by comparing the probability of an object fitting the data's stochastic process.
- Statistical methods for outlier detection are valid when data meets statistical assumptions.

Nonparametric methods

- High-dimensional data presents challenges for statistical outlier detection.
- Computational costs of statistical methods vary with model complexity, being more efficient for simple parametric models and less so for complex models like mixture models and kernel density estimation.

Proximity-based approaches

- Distance measures assess the similarity between objects in feature space, with far objects often labeled as outliers.
- Proximity-based approaches consider an outlier's distance from its nearest neighbors, deviating significantly from most other objects in the dataset.
- Proximity-based outlier detection methods include distance-based and density-based approaches.

Proximity-based approaches

- Distance-based methods define a neighborhood using a given radius and identify an object as an outlier if its neighborhood lacks enough other points.
- Density-based methods analyze an object's density and that of its neighbors, identifying an object as an outlier if its density is considerably lower than its neighbors.

Distance-based outlier detection

- Proximity-based outlier detection relies on distance-based outliers.
- Users set a distance threshold to define object neighborhoods.
- To identify a specific type of outlier, one checks the fraction of objects within an object's neighborhood.
- In this method, an object is considered an outlier if this fraction is below a certain threshold (π).

Distance-based outlier detection

- Alternatively, one can determine outliers by comparing an object's distance to its k -nearest neighbor (k is determined by π).
- Calculating these outliers is achieved through nested loops, which, despite a worst-case time complexity of $O(n^2)$, often run in nearly linear time due to early loop termination for most normal objects.

Density-based outlier detection

- Distance-based outliers are a type of outlier detection method that assesses how far an object is from the majority of other objects in the dataset.
- Proximity-based approaches consist of two main categories: distance-based and density-based methods.
- Distance-based methods determine if an object is an outlier based on its distance to its nearest neighbors within a defined radius.

Density-based outlier detection

- Density-based methods focus on an object's local neighborhood and compare its density to that of its neighbors to identify outliers.
- Distance-based outliers are considered "global outliers" because they rely on global parameters like distance threshold (r) and fraction threshold (π).
- Some datasets exhibit complex structures where local outliers are more relevant, such as objects that deviate from local clusters.

Density-based outlier detection

- Density-based outlier detection methods use the relative density of an object compared to its neighbors to identify local outliers.
- The k-distance of an object measures the distance to its k-nearest neighbor, and the k-distance neighborhood includes all objects with distances less than or equal to the k-distance.
- The reachability distance helps smooth the local density measure, and the local reachability density is the average reachability distance of an object to its neighbors.

Density-based outlier detection

- The local outlier factor (LOF) quantifies the degree to which an object is considered an outlier by comparing its local reachability density to that of its neighbors.
- LOF values are close to 1 for objects within consistent clusters, ensuring they are not mislabeled as outliers.
- LOF is easily interpretable and can identify local outliers with relatively low local density compared to their neighbors.

Reconstruction-based approaches

- Reconstruction-based outlier detection relies on succinctly representing normal data for accurate reconstruction and identifying outliers based on poor reconstruction.
- Key questions include finding the succinct representation, using it for reconstruction, and measuring the reconstruction quality.
- Two methods are matrix-factorization for numerical data and pattern-based compression for categorical data.

Matrix factorization–based methods for numerical data

- Matrix factorization-based outlier detection uses low-rank matrices to represent the original data matrix, allowing the reconstruction of data samples.
- The reconstruction error of data samples indicates their outlierness, with higher errors suggesting outliers.
- The approach includes representing data samples as a matrix, approximating it with low-rank matrices F and G , and computing the reconstruction error for each sample.

Matrix factorization–based methods for numerical data

- Singular value decomposition (SVD) is a common method to find F and G , while example-based matrix factorization reduces redundancy and improves efficiency.
- Nonnegative matrix factorization (NMF) and nonnegative residual matrix factorization (NrMF) are used when data is nonnegative, making the results more interpretable.
- Robust PCA aims to create a sparse residual matrix to identify outliers.
- Matrix factorization is linear, but for capturing nonlinear relationships, autoencoders are used to reconstruct data samples and detect outliers based on reconstruction errors.

Pattern-based compression methods for categorical data

- For data with categorical attributes, you can convert the attributes into binary format and apply matrix factorization-based methods to detect outliers.
- An example involves converting categorical attributes like "income," "credit," and "purchase" into binary attributes. The binary data matrix is then approximated using low-rank matrices to detect outliers.
- This method simplifies categorical data into binary form, making the detection process computationally intensive and results hard to interpret when there are many possible values.

Pattern-based compression methods for categorical data

- An alternative approach uses pattern-based compression methods to detect outliers for categorical data.
- A code table is constructed to represent the input data samples using code words, binary codes, and encoding lengths. Each code word corresponds to a pattern of attribute-value pairs.
- The encoding length serves as an indicator of outlierness, where longer lengths indicate higher outlierness.

Pattern-based compression methods for categorical data

- The pattern-based compression approach is more interpretable compared to matrix factorization, as it highlights unusual patterns in the data.
- The Minimum Description Length (MDL) principle is used to search for the optimal code table, where the best code table minimizes the description length of both the model and the data set.
- MDL considers the trade-off between code table complexity (model description) and data encoding costs.
- Longer encoding lengths suggest outlying data samples that are represented by rare, infrequent patterns.

Clustering- vs. classification-based approaches

- Clustering-based methods that are unsupervised and focus on detecting outliers based on their relationship with clusters.
- Classification-based methods that are supervised and treat outlier detection as a classification problem, training models to differentiate between normal data and outliers.

Clustering-based approaches

- Three primary approaches to clustering-based outlier detection:
 - Identifying objects that do not belong to any cluster.
 - Determining outliers based on the distance to their nearest cluster.
 - Detecting small or sparse clusters and marking objects within them as outliers.

Clustering-based approaches

- Clustering-based methods typically compare individual objects to clusters to identify outliers.
- Some outliers may form small clusters and can be missed by these methods.
- The FindCBLOF algorithm addresses this issue by considering both cluster size and similarity to clusters.

Clustering-based approaches

- Clustering-based methods are computationally efficient and work without labeled data.
- Clustering method choice affects the effectiveness of outlier detection.
- Clustering-based methods may face challenges with large datasets.

Classification-based approaches

- Outlier detection using classification methods faces challenges due to imbalanced training sets.
- Classification-based outlier detection often employs a one-class model, focusing on the "normal" class.
- Methods like one-class SVM and Support Vector Data Description (SVDD) build models for the "normal" class.
- These models define a decision boundary for the "normal" class to distinguish outliers.

Classification-based approaches

- The advantage is the ability to detect new outliers beyond those in the training set, as long as they fall outside the decision boundary.
- This approach can be extended to situations where multiple "normal" classes exist, each with its model.
- Combining classification-based and clustering-based methods in a semi-supervised way is possible.
- Human domain knowledge can be incorporated into the process to enhance the classification model.

Classification-based approaches

- Classification-based methods have a fast detection process and depend on training data quality.
- Active learning can help improve classification-based outlier detection when representative training data is limited.

Mining contextual and collective outliers

- Contextual outliers, also known as conditional outliers, deviate significantly within a specific context defined by contextual attributes.
- Contextual attributes are determined by the application and are often provided by users for the task of contextual outlier detection.
- These attributes can encompass spatial features, time-related information, network locations, and complex structured attributes.
- Behavioral attributes define the characteristics of an object and play a crucial role in assessing whether the object is an outlier within its context.

Transforming contextual outlier detection to conventional outlier detection

- In cases where clear contexts exist, these methods transform contextual outlier detection into standard outlier detection.
- The process involves two steps: first, identifying the context of a data object using contextual attributes, and second, calculating the outlier-ness score within that context using traditional outlier detection techniques.
- For instance, in customer-relationship management, contextual outliers within customer groups can be detected based on attributes like `age_group` and `postal_code` (contextual) and `number_of_transactions_per_year` and `annual_total_transaction_amount` (behavioral).

Transforming contextual outlier detection to conventional outlier detection

- Generalizing contexts is necessary when there are very few or no customers in specific groups. It assumes that customers with similar age and location exhibit similar behavior.
- Mixture models like U and V are learned from training data to represent contextual and behavioral attributes. A mapping, $p(V_i | U_j)$, captures the probability of a data object belonging to context U_j and being generated by cluster V_i . The outlier-ness score is then calculated, transforming contextual outlier detection into outlier detection using mixture models.

Modeling normal behavior with respect to contexts

- In cases where clear data partitioning into contexts is inconvenient or infeasible, contextual outlier detection focuses on modeling normal behavior with respect to contexts.
- Using a training data set, these methods train a model that predicts expected behavior attribute values based on contextual attribute values.
- To identify contextual outliers, the model is applied to the contextual attributes of a data object, and significant deviations in behavior attribute values from the predicted values indicate a contextual outlier.

Modeling normal behavior with respect to contexts

- These methods use prediction models to link contexts and behavior, eliminating the need for explicit context specification.
- Techniques like regression, Markov models, and finite state automaton can be employed to build such models.

Modeling normal behavior with respect to contexts

- Contextual outlier detection enhances conventional outlier detection by considering contexts, helping detect outliers that might otherwise go unnoticed and reducing false alarms.
- For example, it can identify a low-income credit card user with spending patterns similar to millionaires as a contextual outlier, correcting false alarms based on the context.

Mining collective outliers

- Collective outliers are groups of data objects that collectively deviate from the entire data set.
- Detecting collective outliers involves examining data structures, making it more complex.
- Methods for collective outlier detection can reduce the problem to conventional outlier detection by identifying and treating structure units as data objects.

Mining collective outliers

- Another approach directly models the expected behavior of structure units.
- Collective outlier detection is challenging and computationally intensive but offers practical value and room for further development.

Outlier detection in high-dimensional data

- Outlier detection in high-dimensional data is challenging due to the curse of dimensionality.
- Conventional methods relying on proximity or density lose effectiveness as dimensionality increases.
- Ideal methods for high-dimensional data should provide interpretations of outliers, handle data sparsity, model outliers in relevant subspaces, and scale with dimensionality.
- Approaches to high-dimensional outlier detection include extending conventional methods, finding outliers in subspaces, using outlier detection ensembles, applying deep learning, and modeling high-dimensional outliers.

Extending conventional outlier detection

- One approach to high-dimensional outlier detection extends conventional methods.
- It employs proximity-based models but uses alternative measures or subspaces to mitigate high-dimensional deterioration.
- HilOut is an example of this approach, ranking objects based on distance and outputting top outliers.
- To address scalability issues, HilOut uses space-filling curves for approximation in high-dimensional spaces.

Extending conventional outlier detection

- Another approach reduces high-dimensional outlier detection to lower dimensions using dimensionality reduction methods like principal components analysis (PCA).
- PCA can extract a lower-dimensional space, emphasizing principal components with low variance to distinguish normal instances from outliers.
- Extending conventional methods has limitations: it can't detect outliers with respect to subspaces and requires the existence of a well-separated lower-dimensional space.

Finding outliers in subspaces

- Another approach for high-dimensional outlier detection focuses on subspaces.
- Finding outliers in lower-dimensional subspaces can provide valuable context for interpreting why an object is an outlier.
- A method like grid-based subspace outlier detection can be used to identify sparse areas within subspaces.
- Projections onto subspaces are considered, and significantly sparse areas may contain outliers.

Finding outliers in subspaces

- Sparsity coefficients are calculated for regions in subspaces, measuring how many objects are in a region relative to the expected number.
- A cube with a very small sparsity coefficient suggests outliers in the subspace.
- A brute-force search for cubes in all possible subspaces is costly, but more efficient methods like evolutionary searches or genetic algorithms can be used to find outliers in subspaces.
- Detecting outliers in subspaces helps with interpretation but requires addressing efficiency and scalability challenges.

Outlier detection ensemble

- Ensemble methods are effective for high-dimensional outlier detection.
- The process involves creating K base detectors, each operating in a random subspace of lower dimensionality than the original feature space.
- Each base detector assigns outlier-ness scores to data tuples using an off-the-shelf method.
- These scores are aggregated across all base detectors to obtain overall outlier-ness scores for all data tuples.

Outlier detection ensemble

- Data tuples with the highest overall outlier-ness scores are flagged as outliers.
- Two key issues in outlier detection ensemble are how to represent data in a random subspace and how to aggregate detection results.
- Common methods for representing data in a random subspace include feature bagging and rotated bagging.
- Aggregation can be done through mean or max aggregation, and it's important to normalize outlier-ness scores to prevent one base detector from dominating the results.

Taming high dimensionality by deep learning

- Deep learning methods (e.g., autoencoders, neural networks) create low-dimensional embeddings to handle high-dimensionality and complex interactions in data.
- Two strategies for using deep learning in high-dimensional outlier detection are discussed.
- The first strategy uses deep learning to preprocess data, generating lower-dimensional embeddings, which are then processed by standard outlier detection methods.

Taming high dimensionality by deep learning

- The second strategy integrates embedding learning and outlier detection, often leading to improved accuracy. It replaces components of traditional outlier detection methods with deep learning models.
- Examples include OC-NN, which replaces hyperplane output with neural networks, and DevNet, which uses deep learning for outlier-ness scores, outperforming alternative methods by replacing raw feature values.

Modeling high-dimensional outliers

- Some outlier detection methods for high-dimensional data create new models specifically designed for high-dimensional outliers.
- These models often utilize new heuristics that avoid issues associated with proximity measures in high-dimensional data.
- An example of such an approach is Angle-Based Outlier Detection (ABOD).

Modeling high-dimensional outliers

- ABOD examines the angles between data points to assess their variance. Points that deviate from the angle variance are considered outliers.
- Distance-weighted angle variance is used to compute an outlier-ness score for each point.
- The points in the dataset are then ranked in ABOF-ascending order based on their scores.

Modeling high-dimensional outliers

- Exact computation of ABOF for every point in a database is computationally expensive ($O(n^3)$), making it impractical for large datasets. Approximation methods have been developed for efficiency.
- While developing native models for high-dimensional outliers can be effective, finding suitable heuristics and ensuring efficiency and scalability on large high-dimensional datasets remain challenging.

Summary

- Outliers deviate significantly from the rest of the data.
- Types of outliers include global, contextual, and collective.
- Challenges involve data models, application-specific systems, noise differentiation, and justification.
- Methods are supervised, semisupervised, or unsupervised; they can be statistical, proximity-based, or reconstruction-based.

Summary

- Statistical methods use statistical models (parametric or nonparametric).
- Proximity-based methods rely on object proximity or density.
- Clustering-based methods distinguish normal from sparse clusters.
- Classification-based methods utilize one-class models.

Summary

- Contextual and collective methods explore structures using contextual attributes or implicit mining.
- High-dimensional data methods encompass extending conventional, subspace-based, ensemble, deep learning, and native outlier modeling.